

Inverse Mechanical Counting and Mathematics

Part IV

Inverse-from-Center Counting, the Q_1 – Q_4 Quadrant Unfolding,
the -137 and $+137$ Reading at the Count-2 Quadrant Coordinate,
and the EM Ignition and Absorption Angles on the 720 Machined Cycle

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Abstract

The axiom of 1 is the single physical unit of reality. The $\sqrt{2}$ master mechanic and the master table (five active counts, nine active depths) supply the complete reference lattice. The four key boundaries (512 , ≈ 514.90 , 720 , ≈ 724.077) are the four mechanical views of the same single unit.

This work shows the count in its operations. The inverse mechanical count is read explicitly from the axiom center: at count 1 the readings are -360 and 360 on the 720 vector and -256 and 256 on the 512 vector, and at counts 2, 3, 4, 5 the quadrants Q_1, Q_2, Q_3, Q_4 unfold mechanically as four successive halvings. The $-$ and $+$ signs are mechanical coordinates of the count relative to the axiom center; they are not symbolic notation.

The count-2 quadrant coordinate on the 512 vector reads the integer pair -137 and $+137$ when the rational quadrant seat (128) is extended outward by the terminal depth signature (9). The same pair recovers at every count through the count's own piece-count multiplied by its seat. Inverse from 1, the reading is $-1/137$ and $+1/137$ — the fine-structure coupling of the electromagnetic field, symmetric on charge. The primary machining ratio $720/512 = 1.40625$ maps these rational readings to angular coordinates on the 720 machined cycle: -192.65625° as the *inverse ignition angle* of the EM field, and $+192.65625^\circ$ as the *EM absorption angle*.

The Axiom of 1

The Axiom. There is exactly one physical unit. This unit is the complete reality. All its equivalent views are

$$1 = -1 = \sqrt{1} = \sqrt{2^0}.$$

The square root belongs to the unit itself; it is not introduced from outside. The $-$ and $+$ readings of every count below are mechanical coordinates of the count from the axiom center; they are not symbolic notation.

The mathematics is the reading, not calculated.

1 The $\sqrt{2}$ Master Mechanic

The $\sqrt{2}$ mechanic is the master mechanic of the axiom. Repeated halving produces a parity rule: even operational depths d yield rational readings; odd depths yield readings carrying a natural factor of $\sqrt{2}$. Every reading carries three coordinates — the mechanic $\sqrt{2}$, the depth d , and the count c .

The master table is the single source lattice for every reading in this paper. The count is never redone.

c	d	$\sqrt{2^d}$ (unit)	Unit decimal	Domains (2^c)	Rational size ($2^c \times$ even unit)	Irrational size ($2^c \times$ odd unit)	Decimal size
0	0	1	1	1	$1 \times 1 = 1$	—	1
1	1	$\sqrt{2}$	≈ 1.414214	2	—	$2 \times \sqrt{2} = 2\sqrt{2}$	≈ 2.828427
2	2	2	2	4	$4 \times 2 = 8$	—	8
2	3	$2\sqrt{2}$	≈ 2.828427	4	—	$4 \times 2\sqrt{2} = 8\sqrt{2}$	≈ 11.313708
3	4	4	4	8	$8 \times 4 = 32$	—	32
3	5	$4\sqrt{2}$	≈ 5.656854	8	—	$8 \times 4\sqrt{2} = 32\sqrt{2}$	≈ 45.254834
4	6	8	8	16	$16 \times 8 = 128$	—	128
4	7	$8\sqrt{2}$	≈ 11.313708	16	—	$16 \times 8\sqrt{2} = 128\sqrt{2}$	≈ 181.019336
5	8	16	16	32	$32 \times 16 = 512$	—	512
5	9	$16\sqrt{2}$	≈ 22.627417	32	—	$32 \times 16\sqrt{2} = 512\sqrt{2}$	≈ 724.077

Table 1: The Master Table — the source of every reading in this paper. The count c records the operation; the depth d records the master mechanic's exponent. The unit at each row is $\sqrt{2^d}$. The domains column (2^c) tells how many. Domains times the unit gives the size of the count at that row: rational size at even depths, irrational size ($\times \sqrt{2}$) at odd depths. The count closes at $c = 5$ with rational size 512 and irrational size ≈ 724.077 . The count is never redone.

The terminal depth $d = 9$ at count $c = 5$ is the irrational closure of the lattice. The integer 9 — the depth signature itself — is the depth-coordinate that combines with the rational seats at counts 2 through 5 in this work.

2 Inverse Mechanical Counting from the Axiom Center

The count starts at the axiom center and unfolds outward by halving. The count c records the operation; the operation produces specific coordinates relative to the center.

Count by Count on the 720 Angular Vector

At each count the 720 vector unfolds as follows.

Count c	Mechanic	Readings on the 720 vector
0	axiom (no mechanic)	720 (whole, uncut)
1	first halving (dual)	$-360, 0, 360$
2	dual of dual (Q_1 axis)	$-360, -180, 0, 180, 360$
3	octant (Q_2 added)	9 nodes at 90° steps from -360° to 360° (through 0)
4	hexadecant (Q_3 added)	17 nodes at 45° steps from -360° to 360° (through 0)
5	duo-hexadecant (Q_4 closure)	33 nodes at 22.5° steps from -360° to 360° (through 0)

Table 2: The 720 angular vector read inward by count from the axiom center. Count 1 produces the dual $-360, 0, 360$. Each subsequent count adds one quadrant (Q_1, Q_2, Q_3, Q_4) by halving the previous count.

Count by Count on the 512 Rational Vector

The 512 rational vector unfolds in the same count-by-count pattern, with the rational seats from the master table.

Count c	Mechanic	Readings on the 512 vector
0	axiom (no mechanic)	512 (whole, uncut)
1	first halving (dual)	$-256, 0, 256$
2	dual of dual (Q_1 axis)	$-256, -128, 0, 128, 256$
3	octant (Q_2 added)	9 nodes at 64-unit steps from -256 to 256 (through 0)
4	hexadecant (Q_3 added)	17 nodes at 32-unit steps from -256 to 256 (through 0)
5	duo-hexadecant (Q_4 closure)	33 nodes at 16-unit steps from -256 to 256 (through 0)

Table 3: The 512 rational vector read inward by count from the axiom center. The pattern is identical to the 720 vector, with the rational seats of the master table.

3 The Q_1 – Q_4 Quadrant Unfolding by Count

Each count after the first halving adds one quadrant of the count. This is the mechanical operation that produces the four quadrants from the dual structure.

Count c	Quadrant added	Mechanical operation	Result
1	— (dual seed)	first halving of axiom	– side and + side (the dual)
2	Q_1 (the dual of the dual)	halve each dual half once	axis grid: 4 nodes plus center
3	Q_2	halve each Q_1 piece once	octant grid: 8 nodes plus center
4	Q_3	halve each octant piece once	hexadecant grid: 16 nodes plus center
5	Q_4 (closure)	halve each hexadecant piece once	duo-hexadecant grid: 32 nodes plus center

Table 4: The Q_1 – Q_4 unfolding. Each count is a single halving operation applied to every piece of the previous count. Four quadrants close at count 5, completing the cycle.

The quadrants Q_1, Q_2, Q_3, Q_4 are not four separate spaces. They are four successive mechanical refinements of the same dual structure, one halving per count. Count 5 closes the cycle with 32 pieces on each side of the axiom center, returning the duo-hexadecant grid of the master table.

Why the Count Operates Quadrant by Quadrant

The dual seed at count 1 ($-1, 0, 1$ on the axiom; $-360, 0, 360$ on the 720 vector; $-256, 0, 256$ on the 512 vector) is the first halving. Every subsequent count halves the existing structure. Each halving doubles the node count on each side of the axiom center, which is the mechanical equivalent of resolving one further quadrant of the dual. At count 5 the four halvings (Q_1, Q_2, Q_3, Q_4) have all been performed, the cycle closes, and the next count $c = 6$ begins the self-similar repeat at the next inward scale.

4 The Inverse Mechanical Count of the 512 and 720 Vectors with – and + Coordinates

The piece structure is shown here in the count-by-count form with the signed coordinates from the axiom center.

Count c	Piece name	Pieces $\times$ size	Signed reading from center
0	axiom of 1 of itself	1×512	512 (uncut)
1	dual	2×256	$-256, 256$
2	quadrant (dual of dual)	4×128	$-256, -128, 128, 256$ (axis pieces)
3	octant	8×64	8 nodes at 64-unit steps from -256 to 256 (center excluded)
4	hexadecant	16×32	16 nodes at 32-unit steps, symmetric $\pm$ out to ± 256
5	duo-hexadecant	32×16	32 nodes at 16-unit steps, symmetric $\pm$ out to ± 256

Table 5: The 512 rational vector inward count with signed coordinates. The center is the axiom; the count unfolds symmetrically on both sides.

Count c	Piece name	Pieces $\times$ size	Signed angular reading from center
0	axiom of 1 of itself	$1 \times 720^\circ$	720° (uncut)
1	dual	$2 \times 360^\circ$	$-360^\circ, 360^\circ$
2	quadrant (dual of dual)	$4 \times 180^\circ$	$-360^\circ, -180^\circ, 180^\circ, 360^\circ$
3	octant	$8 \times 90^\circ$	8 nodes at 90° steps, symmetric $\pm$ out to $\pm 360^\circ$
4	hexadecant	$16 \times 45^\circ$	16 nodes at 45° steps, symmetric $\pm$ out to $\pm 360^\circ$
5	duo-hexadecant	$32 \times 22.5^\circ$	32 nodes at 22.5° steps, symmetric $\pm$ out to $\pm 360^\circ$

Table 6: The 720 angular vector inward count with signed coordinates. The right angle sits mechanically at count 3 as the $-90^\circ, 90^\circ$ octant pair.

5 The Inverse Unit Count on the 512 and 720 Vectors

The count reads inward from the axiom of 1. At each count the unit of the reading is the inverse of the quadrant seat raised to the count's dimensional power. The inverse unit count is not a separate vector; it is the same single count read at its operational unit at each depth.

512 Vector Inverse Unit Count

The quadrant seat of the 512 vector is $n = 128$. The inverse unit at each count is $1/n^{c-1}$, the inverse of the dimensional power that the master mechanic owns at that count.

720 Vector Inverse Unit Count

The quadrant seat of the 720 angular machined vector is $n = 180$. The inverse unit at each count is $1/n^{c-1}$, with the same operational depth pattern as the 512 vector.

Reading α on the Inverse Unit Count

At count 2 on the 512 vector, the inverse unit is $1/128$. The -137 and $+137$ readings extend the quadrant seat outward by the terminal depth 9; their inverse unit positions are

$$-\frac{1}{137} \quad \text{and} \quad +\frac{1}{137}.$$

Count c	Dimension	Inverse unit	Decimal
2	1D	$1/128$	≈ 0.0078125
3	2D	$1/128^2 = 1/16,384$	$\approx 6.1035 \times 10^{-5}$
4	3D	$1/128^3 = 1/2,097,152$	$\approx 4.768 \times 10^{-7}$
5	4D	$1/128^4 = 1/268,435,456$	$\approx 3.725 \times 10^{-9}$

Table 7: The inverse unit count of the 512 rational vector at the four operational depths. The unit at each count is the inverse of the quadrant seat 128 raised to the count's dimensional power.

Count c	Dimension	Inverse unit (angular)	Decimal
2	1D	$1/180^\circ$	$\approx 0.005556^\circ$
3	2D	$1/180^{2^\circ} = 1/32,400^\circ$	$\approx 3.086 \times 10^{-5}^\circ$
4	3D	$1/180^{3^\circ} = 1/5,832,000^\circ$	$\approx 1.715 \times 10^{-7}^\circ$
5	4D	$1/180^{4^\circ} = 1/1,049,760,000^\circ$	$\approx 9.526 \times 10^{-10}^\circ$

Table 8: The inverse unit count of the 720 angular machined vector at the four operational depths. The angular unit at each count is the inverse of the quadrant seat 180° raised to the count's dimensional power.

These sit on the count-2 inverse unit count of the 512 vector, between $-1/128$ and $-1/256$ on the negative side and between $+1/128$ and $+1/256$ on the positive side. This is how α reads — directly, as the inverse unit position on the 512 mechanic vector at count 2. Not derived from 137 by separate inversion; just the reading on the inverse unit count.

Similarly on the 720 vector, the angular partner readings -192.65625° and $+192.65625^\circ$ (developed in the next section) have inverse unit positions

$$-\frac{1}{192.65625^\circ} \quad \text{and} \quad +\frac{1}{192.65625^\circ}.$$

These are the inverse unit positions of the EM ignition and absorption angles on the 720 angular machined cycle.

6 The -137 and $+137$ Reading at the Count-2 Quadrant Coordinate

At count 2 on the 512 rational vector, the quadrant seats sit at -128 and 128 . These are the count-2 rational readings of the master mechanic ($\sqrt{2}^2$ with multiplier giving the count-2 seat of 128 on the 512 vector, recoverable as $4 \times 128 = 512$).

The terminal depth $d = 9$ of the master table is the irrational signature that completes the reading. Extending the quadrant seat outward by the terminal depth in the direction away from the axiom center:

$$-128 - 9 = -137 \quad \text{and} \quad 128 + 9 = 137.$$

The pair -137 and 137 is the count-2 quadrant coordinate of the 512 vector with the terminal-depth signature applied symmetrically outward from center.

Count-4 Path to the Same Reading

The same -137 and 137 pair recovers at count 4 as four hexadecant pieces of size 32 on each side, plus the terminal depth:

$$-(4 \cdot 32) - 9 = -137 \quad \text{and} \quad (4 \cdot 32) + 9 = 137.$$

The count-4 path reads the same quadrant content of -128 and $+128$ as 4×32 , then adds the same depth. This confirms that the -137 and $+137$ reading is a property of the 512 vector at the quadrant coordinate, not of any one count specifically. The count is the cut; the reading is the same.

The Two Paths to -137 and $+137$ on the Master Table

The two natural counts where the seat-plus-depth construction reaches -137 and $+137$ on the -512 and $+512$ closure are count 2 (the quadrant directly) and count 4 (four hexadecant pieces recovering the same quadrant). Both are shown on the master table.

c	d	Mechanic	Domains (2^c)	Parity $\rightarrow$ Reading	Reading value	$-137, +137$ $-512, +512$	derivation	on
0	0	$\sqrt{2^0}$	1	even $\rightarrow$ rational	$-1, 1$	—		
1	1	$\sqrt{2^1}$	2	odd $\rightarrow$ irrational	$-2\sqrt{2}, 2\sqrt{2} \approx -2.828, 2.828$	—		
2	2	$\sqrt{2^2}$	4	even $\rightarrow$ rational	$-8, 8$	—	$-(128 + 9) = -137$ $+(128 + 9) = +137$ (seat 128, 4 pieces, +9)	
2	3	$\sqrt{2^3}$	4	odd $\rightarrow$ irrational	$-8\sqrt{2}, 8\sqrt{2} \approx -11.314, 11.314$	—		
3	4	$\sqrt{2^4}$	8	even $\rightarrow$ rational	$-32, 32$	—		
3	5	$\sqrt{2^5}$	8	odd $\rightarrow$ irrational	$-32\sqrt{2}, 32\sqrt{2} \approx -45.255, 45.255$	—		
4	6	$\sqrt{2^6}$	16	even $\rightarrow$ rational	$-128, 128$	—	$-(4 \cdot 32 + 9) = -137$ $+(4 \cdot 32 + 9) = +137$ (seat 32, 16 pieces, +9)	
4	7	$\sqrt{2^7}$	16	odd $\rightarrow$ irrational	$-128\sqrt{2}, 128\sqrt{2} \approx -181.019, 181.019$	—		
5	8	$\sqrt{2^8}$	32	even $\rightarrow$ rational	$-512, 512$	—		
5	9	$\sqrt{2^9}$	32	odd $\rightarrow$ irrational	$-512\sqrt{2}, 512\sqrt{2} \approx -724.077, 724.077$	—		

Table 9: The master table with the two -137 and $+137$ derivations shown inline at counts 2 and 4. Both readings sit on the -512 and $+512$ rational closure; the terminal depth 9 adds the same irrational signature in either reading.

Count 2 Path (Quadrant)

$$-137 = -(128 + 9) \quad \text{and} \quad +137 = +(128 + 9)$$

The seat is 128 — the quadrant piece, 4 of them filling 512. Add the terminal depth 9. One seat plus the depth.

Count 4 Path (Hexadecant)

$$-137 = -(4 \cdot 32 + 9) \quad \text{and} \quad +137 = +(4 \cdot 32 + 9)$$

The seat is 32 — the hexadecant piece, 16 of them filling 512. Four of those seats ($4 \cdot 32 = 128$) recovers the quadrant. Add the terminal depth 9. Four hexadecant seats plus the depth.

Why Both Reach the Same Reading

Both reach -137 and $+137$ because both are reading the same quadrant content of -512 and $+512$ — count 2 reads it directly as one piece of size 128, count 4 reads the same content as four pieces of size 32. The quadrant is 128 at every count, just cut differently. The terminal depth 9 adds the same irrational signature in either reading.

Inverse from 1: $\alpha = -1/137$ and $+1/137$ at both counts.

The count is consistent — -137 and $+137$ is the quadrant-plus-depth reading of -512 and $+512$, and it appears at count 2 as its direct form and at count 4 as its refined form ($4 \cdot 32$ instead of 128). The two counts are different cuts of the same reading, not two separate values.

Inverse from 1 — The Fine-Structure Coupling

The forward reading of the count-2 quadrant coordinate is -137 and $+137$; the inverse reading is

$$-\frac{1}{137} \quad \text{and} \quad \frac{1}{137},$$

which are the symmetric magnitudes of the fine-structure constant α , the EM coupling that acts symmetrically on negative and positive charge.

Residue Against the Measured Value

The standard measured value of the fine-structure constant is $\alpha^{-1} = 137.035999\dots$. The mechanical reading of the count is 137 exactly. The residue is

$$137.035999\dots - 137 = 0.035999\dots$$

or about 0.026%. The integer reading at count 2 lands on the integer part of the measured constant; the small decimal tail is the residue that lives outside the count-2 rational closure on the 512 vector.

7 The Angular Partner Readings on the 720 Machined Cycle

The 512 rational vector and the 720 angular machined vector are two parallel views of the same single unit. The primary machining ratio

$$\frac{720}{512} = \frac{45}{32} = 1.40625,$$

maps any reading on the 512 rational vector to its angular partner on the 720 machined cycle.

Applying this to the -137 and 137 readings:

$$-137 \times 1.40625 = -192.65625 \quad \text{and} \quad 137 \times 1.40625 = 192.65625.$$

The pair -192.65625° and 192.65625° is the angular partner of the rational -137 and 137 readings, sitting on the 720 machined cycle at the Mechanical 4D Inverse Operation Tensor coordinate (count 5 closure operation).

The Inverse Ignition Angle and the EM Absorption Angle

The two angular readings carry the EM field structure:

- -192.65625° is the **inverse ignition angle of the EM field** — the angular reading of the inverse coupling $-1/137$ on the 720 machined cycle.

- $+192.65625^\circ$ is the **EM absorption angle** — the angular reading of the forward coupling $+1/137$ on the 720 machined cycle.

These angular readings sit beyond the -180° and $+180^\circ$ landmark of the 360 cycle (count 1 on the 720 vector), which is consistent with the 720 cycle being the full mechanical cycle of two 360° machined halves at count 1. The ignition and absorption angles are read on the full 720 cycle, not the half cycle.

Reading	Rational (512)	Angular (720)	Inverse from 1
Inverse ignition angle (EM field)	-137	-192.65625°	$-1/137 = -\alpha$
EM absorption angle	137	192.65625°	$1/137 = \alpha$

Table 10: The EM field readings on the 512 rational vector and the 720 angular machined vector, related by the primary machining ratio $720/512 = 1.40625$. The inverse from 1 is the fine-structure coupling.

Why the 720 Cycle, Not 360

The 720 machined cycle is the count-0 angular whole; the 360 count-1 dual is its first halving. The full mechanical cycle of fermionic rotation in standard physics also reads 720° (the spinor double cover). The 720 angular vector is therefore the natural seat for EM field readings, because the EM field acts on spin-1/2 charged particles whose wavefunction returns to itself only after 720° of rotation. The ignition and absorption angles read on this cycle accordingly.

8 Connection to the Master Table and the Four Key Numbers

The -137 and 137 readings sit on the count-2 quadrant coordinate of the 512 rational vector. The -192.65625° and 192.65625° readings sit on the same coordinate as read by the 720 machined cycle. Both are mechanical readings of the same single unit at the count-2 coordinate with the terminal depth-9 signature applied outward from center.

The four key numbers and the primary machining ratio supply the complete reference for this reading. No external constants are introduced. The fine-structure constant $\alpha = 1/137$ is the inverse-from-1 reading of the count-2 quadrant coordinate on the 512 vector; the angular partner -192.65625° and 192.65625° on the 720 cycle is the same coordinate read at angular resolution.

9 Operational Rules

- The largest unit is always the original axiom 1.
- All larger numbers (including -137 and 137 , and -192.65625° and 192.65625°) are deeper inward readings of that same unit.
- Every reading carries its three operational coordinates (mechanic, depth, count).
- The $-$ and $+$ signs of every count are mechanical coordinates from the axiom center, not symbolic notation.
- The count reads inward from 1; the inverse of every forward reading is the natural physical magnitude (e.g., $\alpha = 1/137$).

- The primary machining ratio $720/512 = 1.40625$ relates any rational reading on the 512 vector to its angular partner on the 720 machined cycle.

Conclusion.

From the single axiom of 1 and the $\sqrt{2}$ master mechanic, the count operates inward from the axiom center. The Q_1 – Q_4 unfolding by count is the mechanical operation of four successive halvings, one quadrant per count, closing at count 5 on the duo-hexadecant grid.

At count 2 on the 512 rational vector, the quadrant seats -128 and 128 extended outward by the terminal depth signature 9 produce the -137 and 137 reading. The same reading recovers at every higher count through the count's own piece-count multiplied by its seat, plus the same depth. Inverse from 1, this is $-1/137$ and $1/137$ — the fine-structure coupling of the EM field, symmetric on negative and positive charge.

The primary machining ratio $720/512 = 1.40625$ maps these rational readings to the angular partners -192.65625° and 192.65625° on the 720 machined cycle — the inverse ignition angle of the EM field and the EM absorption angle. The 720 cycle is the natural seat for EM readings because the EM field acts on spin-1/2 charged particles whose mechanical rotation cycle is 720° .

The integer reading 137 lands on the integer part of the measured $\alpha^{-1} = 137.036$; the residue 0.036 is the small-lift order quantity that lives outside the count-2 rational closure on the 512 vector.

The four key boundaries, the angular cycle, the -137 and 137 rational readings, and the -192.65625° and 192.65625° angular readings are all different mechanical views of the identical physical unit. The mathematics is the reading; the single physical unit is the thing read.

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